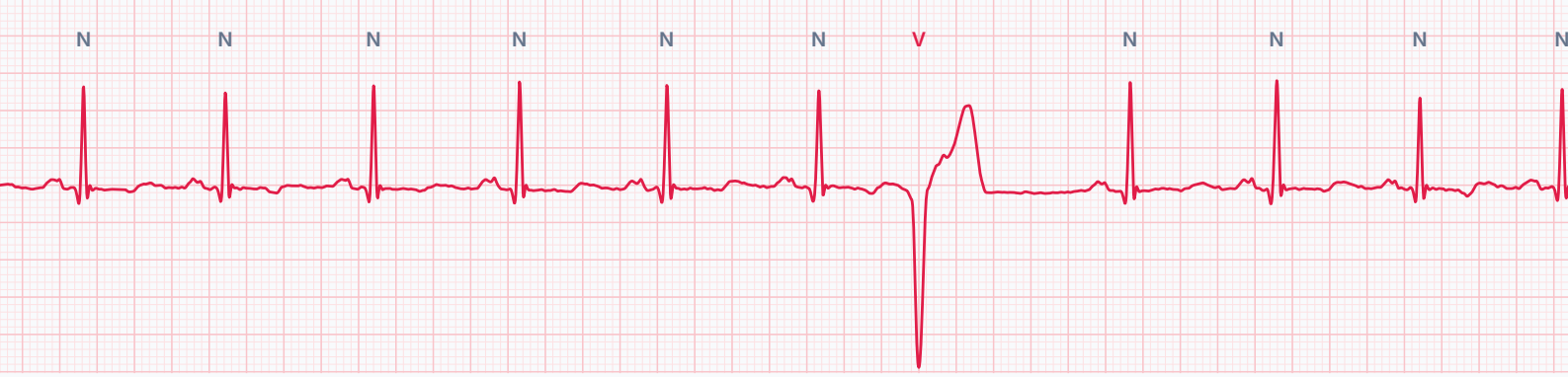


ECG Arrhythmia · project report

MIT-BIH Arrhythmia Database | ANSI/AAMI EC57 | Inter-patient DS1 / DS2 | Built with orchestrated AI agents

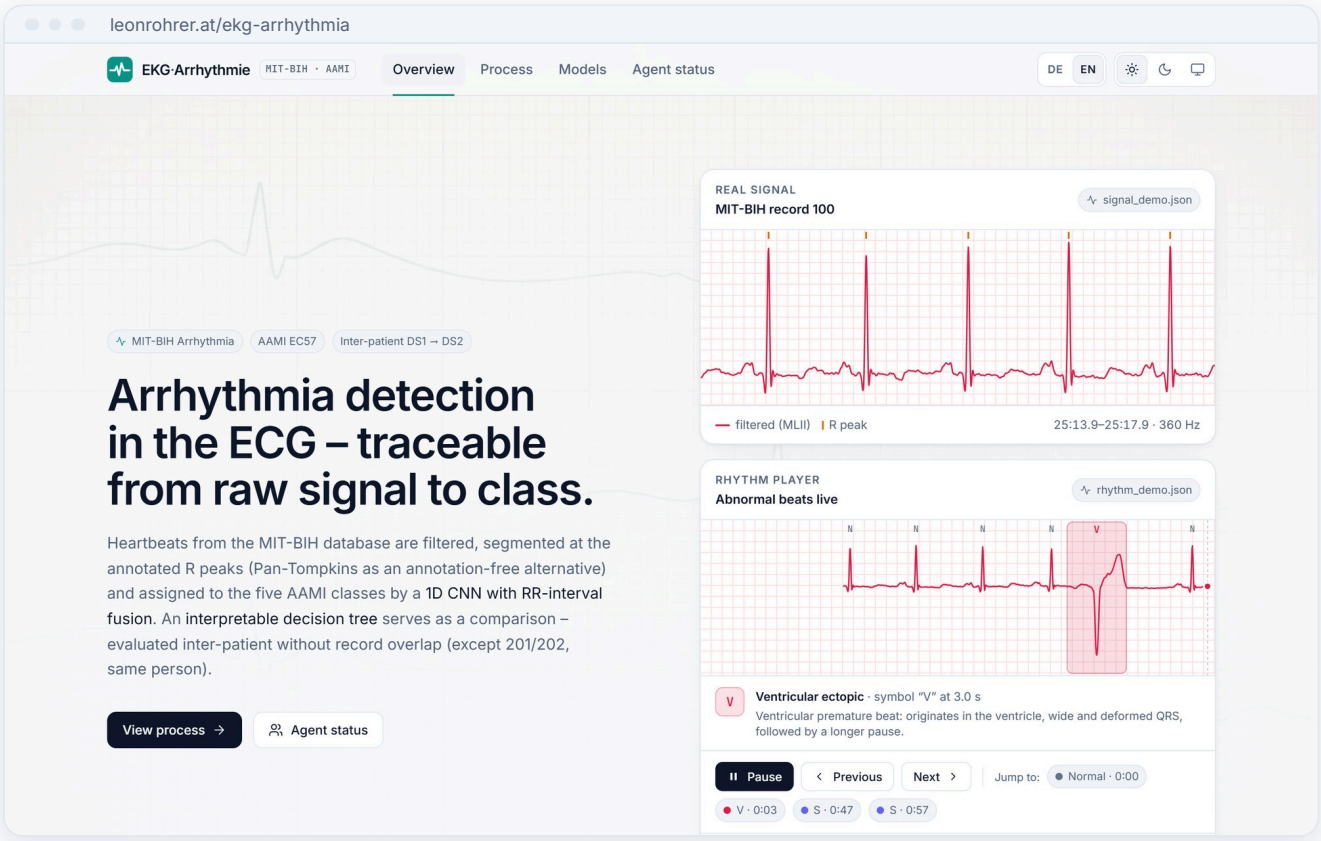
Arrhythmia detection in the ECG — from raw signal to beat class

A complete, reproducible pipeline that sorts every heartbeat of a long-term ECG into the five AAMI classes, with a compact 1D convolutional network, an interpretable decision tree and an honest inter-patient evaluation on patients the models have never seen.

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DATE
3 November 2026

INTERACTIVE VERSION
leonrohrer.at/ekg-arrhythmia



SUMMARY

Finding the rare beats that matter

Every heartbeat of an ECG recording is assigned to one of the five classes of the ANSI/AAMI EC57 standard. The beats that matter clinically — premature beats from the atria (S) or from the ventricles (V) — are rare: 89.0% of all test beats are normal. This project builds the complete chain from the raw signal to the class and evaluates it the strict way, on the records of 22 patients that the models never saw during development.

Problem

Ectopic beats are clinically relevant but hidden among thousands of normal beats. Many published results mix beats of the same patient into training and test data, which inflates the scores far beyond what a new patient would see.

Approach

MIT-BIH Arrhythmia Database, split inter-patient into DS1 (training) and DS2 (test) after de Chazal et al. (2004). Filtering, Pan-Tompkins R-peak detection, one window per beat plus RR intervals; an ensemble of 5 compact 1D CNNs (14,857 parameters each) and an interpretable decision tree.

Built with AI agents

A coordinating “mother agent” split the work into 18 modules for up to 16 parallel sub-agents, held together by binding interface contracts and a lock-safe status protocol. Goals, rules and final checks stayed with the human.

Key result: 1D CNN on the unseen test patients (DS2)

S SUPRAVENTRICULAR BEATS

66.1% sensitivity — **43.0%** PPV

1,837 S beats in the test set

V VENTRICULAR BEATS

97.2% sensitivity — **81.3%** PPV

3,220 V beats in the test set

MACRO-F1

0.59 N, S, V, F — **0.47** all five classes

Accuracy 92.1% is only a side note: answering “normal” every time would already reach 89.0%.

Sensitivity is the share of the true beats of a class that the model found; the **positive predictive value (PPV)** is the share of the model's alarms for that class that were right. For orientation: de Chazal et al. (2004) reported Se 75.9% / PPV 38.5% for S and Se 77.7% / PPV 81.9% for V on the same split (details in chapter 5). The CNN is an ensemble of 5 models; single models reach a macro-F1 of 0.56 ± 0.05 (N, S, V, F). The interpretable decision tree reaches a macro-F1 of 0.36 (N, S, V, F).

Read with care

- The test split DS2 was evaluated twice during development (documented). The results shown are the third and last DS2 evaluation, run once with the frozen final configuration.
- F and Q are practically not recognized (F1 0.01 and 0.00): DS1 has too few and too one-sided examples of them.
- This is a research and portfolio project, not a medical device; it must not be used for diagnosis.

Interactive version with all steps and live charts: https://leonrohrer.at/ekg-arrhythmia/ · Code: https://github.com/LeonTheGoatRohrer/ekg-arrhythmia

1 · BACKGROUND

What the ECG shows — and why evaluation is tricky

The ECG in one minute

The heart is a muscle that contracts about once per second, each time triggered by an electrical impulse from the **sinus node**, its built-in pacemaker. Electrodes on the skin pick up this activity; the voltage over time is the **electrocardiogram (ECG)**.

A normal beat has a typical shape: the small **P wave** (the atria contract), the sharp **QRS complex** (the large ventricles contract) with its tallest spike, the **R peak**, and the broader **T wave** (the ventricles recover). The time between two R peaks is the **RR interval** — at 60 beats per minute exactly one second.

An **arrhythmia** is a disturbance of this rhythm: a beat comes too early, from the wrong place or with the wrong shape. Many are harmless, some are dangerous.

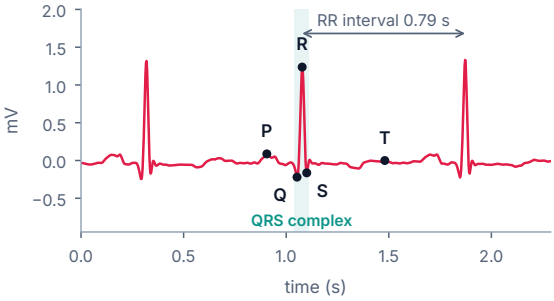


Figure 1. Three normal beats from MIT-BIH record 100 (filtered), with the waves of the middle beat and one RR interval marked.

Five beat classes (ANSI/AAMI EC57)

The MIT-BIH annotations distinguish more than a dozen beat types. To make results comparable, the **ANSI/AAMI EC57** standard groups them into five classes (annotations that do not mark a beat are ignored). S and V carry most of the clinical value — frequent supraventricular ectopic beats can precede atrial fibrillation, frequent or coupled ventricular ones point to structural heart disease — and they are also the rare ones, which makes them hard for algorithms.

Class	Meaning	MIT-BIH symbols	Typical cue
N	Normal and bundle branch block beats	`N, L, R, e, j`	regular timing, normal shape
S	Supraventricular ectopic (from the atria)	`A, a, J, S`	too early; shape often almost normal
V	Ventricular ectopic (from the ventricles)	`V, E`	wide, deformed QRS; pause afterwards
F	Fusion of a normal and a ventricular beat	`F`	shape in between N and V
Q	Paced and unclassifiable beats	`/, f, Q`	almost absent without paced records

Why inter-patient instead of intra-patient

Beats from the same person look very much alike. If all beats are shuffled and split at random (*intra-patient*), the test set contains near-copies of beats the model already knows: it learns "what does patient X look like" instead of "what does a ventricular beat look like". Many published accuracies around 99% rest on such splits — but a new patient is, by definition, unseen. This project therefore follows the **inter-patient** protocol of de Chazal et al. (2004): two disjoint sets of 22 records, DS1 for everything during development and DS2 only for testing.

Intra-patient

train
test

Same patients (colors) in both — the model recognizes people, the score is inflated.

Inter-patient

train
test

Unseen patients — a realistic estimate for a new person.

2 · DATA

MIT-BIH: 44 usable records, 22 + 22 for a fair test

The **MIT-BIH Arrhythmia Database** (Moody & Mark, 2001; distributed by PhysioNet) is the reference dataset of arrhythmia research: 48 half-hour excerpts of two-channel ambulatory ECGs from 47 subjects, recorded at Beth Israel Hospital in Boston, sampled at 360 Hz with 11-bit resolution. At least two cardiologists annotated every single beat independently; disagreements were resolved by consensus. The project uses lead `MLII`, in which the R peaks are particularly clear.

- **Excluded:** the four records with a pacemaker (102, 104, 107, 217), as recommended by AAMI — paced beats have their own shape. 44 records remain.
- **Same person:** records 201 and 202 come from the same subject but sit in different sets — a known, small imprecision of the standard split, accepted for comparability with the literature.

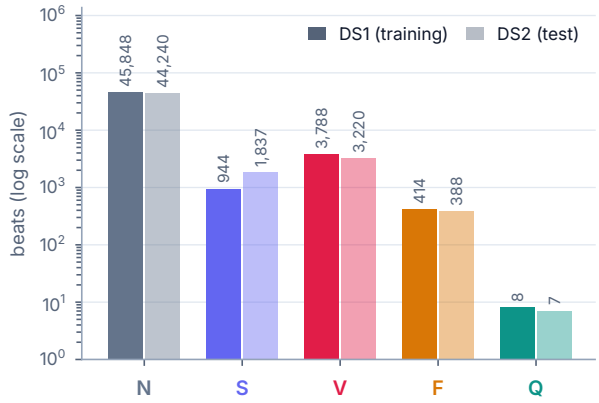


Figure 2. Beats per class and split (dataset_stats.json). The axis is logarithmic: N outnumbers S, F and Q by two to four orders of magnitude.

DS1 · training and validation · 22 records

`101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230`

DS2 · test only · 22 records

`100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234`

Class distribution

Class	DS1 beats	share	DS2 beats	share
N normal	45,848	89.9%	44,240	89.0%
S supraventricular ectopic	944	1.9%	1,837	3.7%
V ventricular ectopic	3,788	7.4%	3,220	6.5%
F fusion	414	0.8%	388	0.8%
Q unclassifiable	8	0.0%	7	0.0%
Total	**51,002**		**49,692**	

Why this matters later

N makes up 89.9% of DS1 and 89.0% of DS2. DS1 contains only 414 fusion (F) beats — about 90% of them from a single record, 208 — and 8 Q beats. Classes this small and this concentrated can hardly be learned in a way that transfers to new patients, and every metric on them is very noisy.

3 · PIPELINE

From raw signal to beat: the pipeline step by step

Seven stations turn a half-hour recording into one class label per heartbeat. All signal figures in this chapter show real data from MIT-BIH record 100 (part of the test set DS2; used here only for illustration, never for training).



3.1 Filtering without shifting the beats

A real ECG is never clean. **Baseline wander** — a slow drift caused mainly by breathing and electrode movement — lies well below 1 Hz; **power-line interference** sits at 60 Hz in this US database; **muscle noise** is broadband and fast. The information that matters for beat classification lies roughly between 0.5 and 40 Hz.

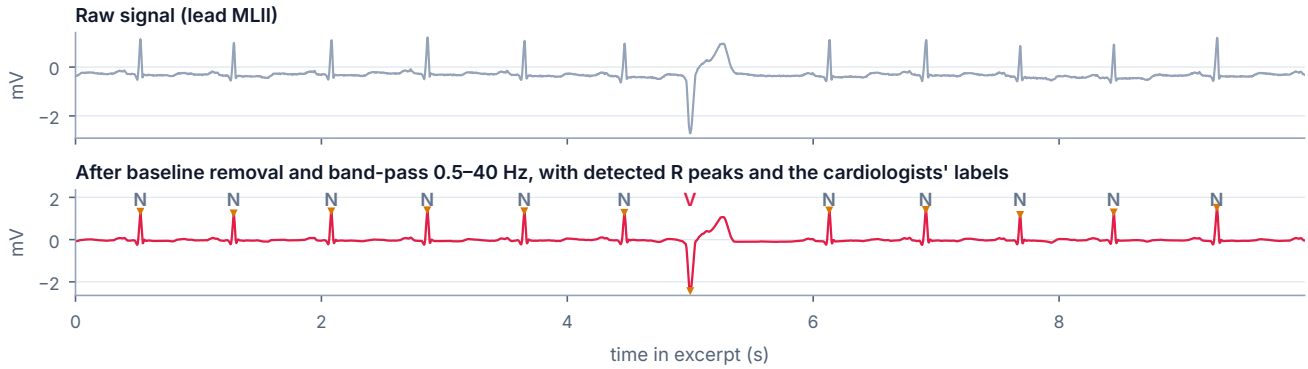


Figure 3. A 10-second excerpt of record 100 before and after filtering. Letters above the beats are the cardiologists' classes; triangles mark the R peaks found by the detector.

Stage 1 – baseline removal (after de Chazal et al., 2004): a 200 ms median filter removes QRS complexes and P waves, a second one with 600 ms also removes the T waves; the remaining baseline estimate is subtracted. Median filters are robust against the steep QRS spikes and do not smear them like a moving average.

Stage 2 – band-pass 0.5–40 Hz: a Butterworth filter of order 4 removes the remaining drift, muscle noise and mains hum. Filtering forward and backward (`sosfiltfilt`) cancels the phase shift, so the R peak stays exactly in place — important because the next steps cut fixed windows around it.

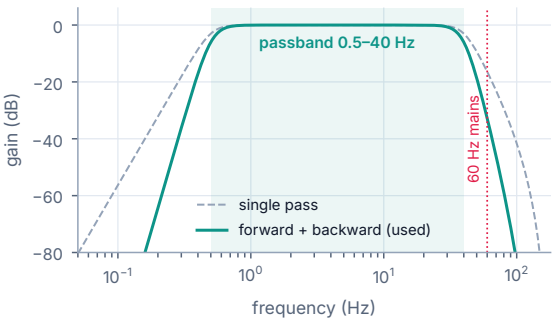


Figure 4. Frequency response of the band-pass (filter_response.json). Filtering forward and backward doubles the attenuation outside the passband.

3.2 Finding every beat: Pan-Tompkins R-peak detection

Before beats can be classified, each one has to be found. The classic algorithm of **Pan & Tompkins (1985)** turns the ECG into a signal in which every QRS complex becomes one clear bump: a 5–15 Hz band-pass emphasizes the QRS and suppresses P and T waves, the derivative highlights steep slopes, squaring makes everything positive and amplifies large slopes, and a 150 ms moving-window integration smooths this into an envelope about as wide as a QRS complex. Adaptive thresholds then decide what is a beat, with a 200 ms refractory period, a T-wave check and a search-back when a beat seems to be missing.

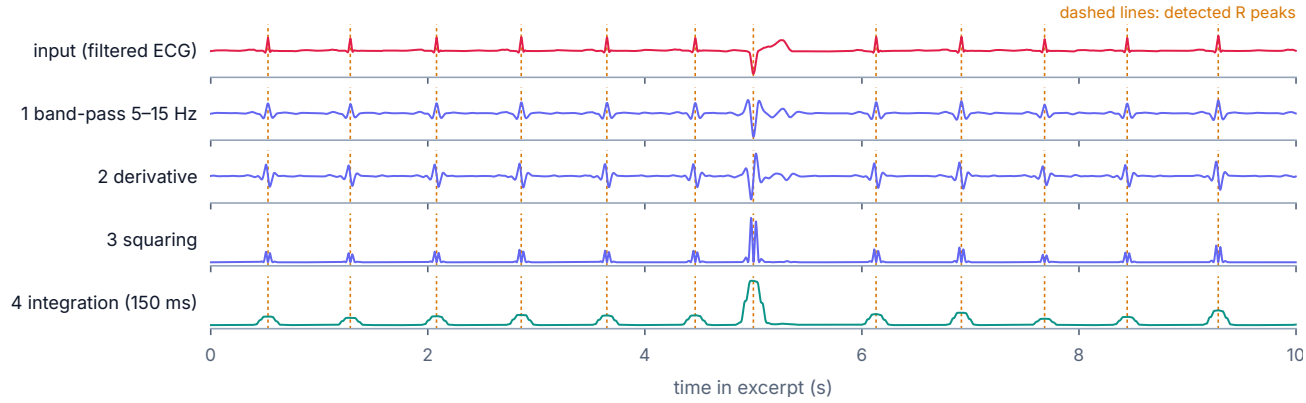


Figure 5. The intermediate stages of the detector on the same excerpt (each row scaled to its maximum).

> **Check on record 100:** 2,273 of 2,273 annotated beats found, 0 false detections (tolerance ±150 ms) — a single, rather clean record, not a database benchmark. On all 22 DS2 records the detector reaches a sensitivity of 99.39% and a PPV of 99.79%. The main results use the cardiologists' reference beat positions, as in de Chazal et al. (2004); chapter 5 adds an end-to-end evaluation on detected beats.

3.3 One window per beat

Around every R peak, a window from **250 ms before** to **450 ms after** the peak is cut out — 252 samples at 360 Hz, R always at the same index. The asymmetric window contains the P wave before the QRS and the longer T wave after it; both carry diagnostic information. Each window gets its class from the cardiologists' annotation and is scaled to mean 0 and standard deviation 1, so the model learns the *shape* rather than the amplitude, which varies with electrode position and body build.

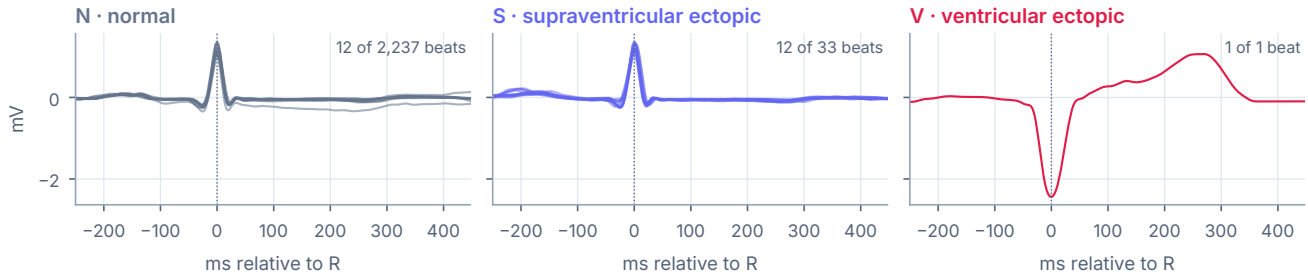


Figure 6. Real beat windows from record 100 (segmentation_demo.json), before z-normalization. The V beat is wide and deformed; the S beats look almost normal. Classes F and Q do not occur in record 100.

3.4 Rhythm context: the timing gives an early beat away

A window shows the shape but not the timing. Four RR features add it: `pre_rr` (to the previous beat), `post_rr` (to the next), `local_rr` (mean of the last up to ten intervals) and the ratio `pre_rr / local_rr`. Ectopic beats come *early*, so the ratio drops well below 1, and a ventricular beat is often followed by a compensatory pause. For S beats, whose shape is often nearly normal, prematurity is frequently the decisive cue; the ratio also keeps it comparable between people with different heart rates.

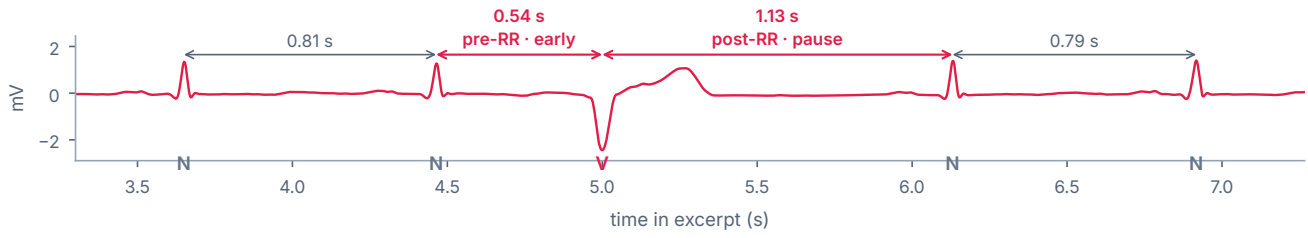


Figure 7. RR intervals around the V beat of the excerpt: pre-RR 0.54 s against a local rhythm of about 0.75 s (ratio ≈ 0.72), followed by a 1.13 s pause.

4 · MODELS

Two models: a compact CNN and an interpretable tree

4.1 A compact 1D convolutional network

A **1D convolutional neural network** slides small learned filters across the beat window. Each filter is a pattern detector that responds where the signal resembles its shape — a steep slope, a wide QRS, an inverted wave. Unlike the features of the decision tree, these patterns are not defined by hand but learned from the data. Three convolution blocks (convolution, batch normalization, ReLU, max-pooling) halve the time resolution step by step, so deeper layers see larger parts of the beat. A global average then condenses each channel into one number. In parallel, the four RR features pass through a small dense layer. Both branches are concatenated and mapped by a small classification head to one score per AAMI class — the network judges *shape and timing together*. With 14,857 trainable parameters it is deliberately small, leaving little room to memorize the 18 training records, and the whole training run took about 6.2 minutes on a CPU.

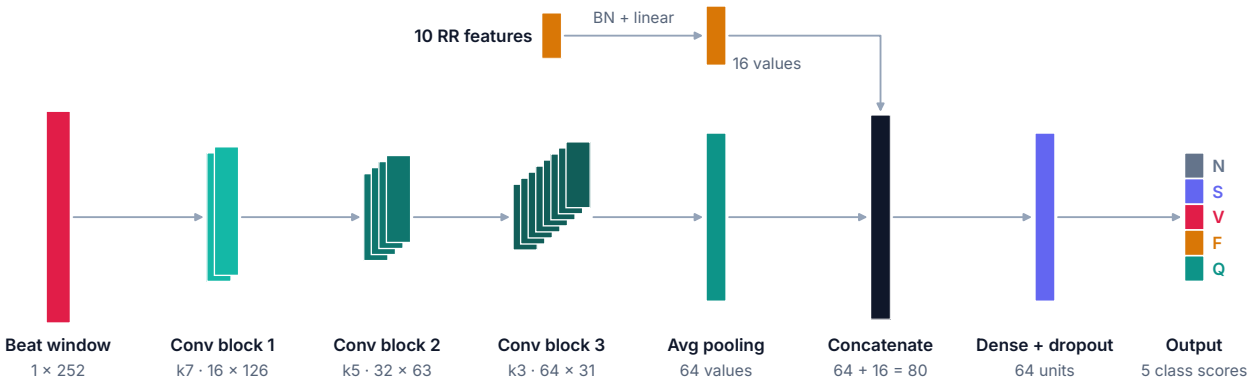


Figure 8. ECGNet1D as drawn from architecture.json: the time axis shrinks from 252 samples while the number of learned channels grows (k = kernel size); the RR branch joins before the classification head.

Stage	Layers	Output shape	Parameters
Beat window (input)	Input	1 × 252	0
RR features (input)	Input	10	0
Conv block 1	Conv1d · BatchNorm1d · ReLU · MaxPool1d	16 × 126	160
Conv block 2	Conv1d · BatchNorm1d · ReLU · MaxPool1d	32 × 63	2,656
Conv block 3	Conv1d · BatchNorm1d · ReLU · MaxPool1d	64 × 31	6,336
Global average pooling	AdaptiveAvgPool1d · Flatten	64	0
RR branch	BatchNorm1d · Linear · ReLU	16	196
Concatenate	Concat	80	0
Dense layer + dropout	Linear · ReLU · Dropout	64	5,184
Output layer	Linear	5	325
Total			**14,857**

4.2 Training against the imbalance

Weighted loss. The cross-entropy loss gives rare classes a larger weight, so a missed S beat costs much more than a missed N beat; without this, the network would largely ignore the rare classes. **Validation inside DS1 only:** held-out DS1 records decide when to stop. DS2 is used neither for choosing settings nor for the stopping

point — otherwise test information would leak into the model. Records that hold a large share of a rare class stay in training: record 208 alone holds about 90% of all DS1 fusion beats.

Class weights	inverse class frequency, capped at 50× the most common class
Optimizer	Adam, learning rate 0.001, batches of 256 beats, up to 20 epochs
Inputs	lead MLII, 4 RR features
Validation	own DS1 records per seed — 0: 114, 119, 122, 205; 1: 116, 118, 205, 230; 2: 106, 112, 114, 205; 3: 106, 109, 112, 201; 4: 119, 124, 215, 220; early stopping after 5 epochs without improvement; records holding at least 40% of a class stay in training
Random seeds	0, 1, 2, 3, 4 — one model per seed, probabilities averaged
Kept epochs	2, 2, 5, 1, 4 (of 7, 7, 10, 6, 9 run)



Figure 9. Training history (metrics.json, model of seed 0). The dotted line marks the epoch whose weights were kept.

4.3 An interpretable decision tree for comparison

The second model is a decision tree of depth 4 with 16 leaves, trained on 51,002 DS1 beats with balanced class weights. Instead of raw samples it receives 11 hand-crafted features that a cardiologist would recognize. Every path from the root to a leaf is a readable if-then rule — the price is far less expressive power than the CNN.

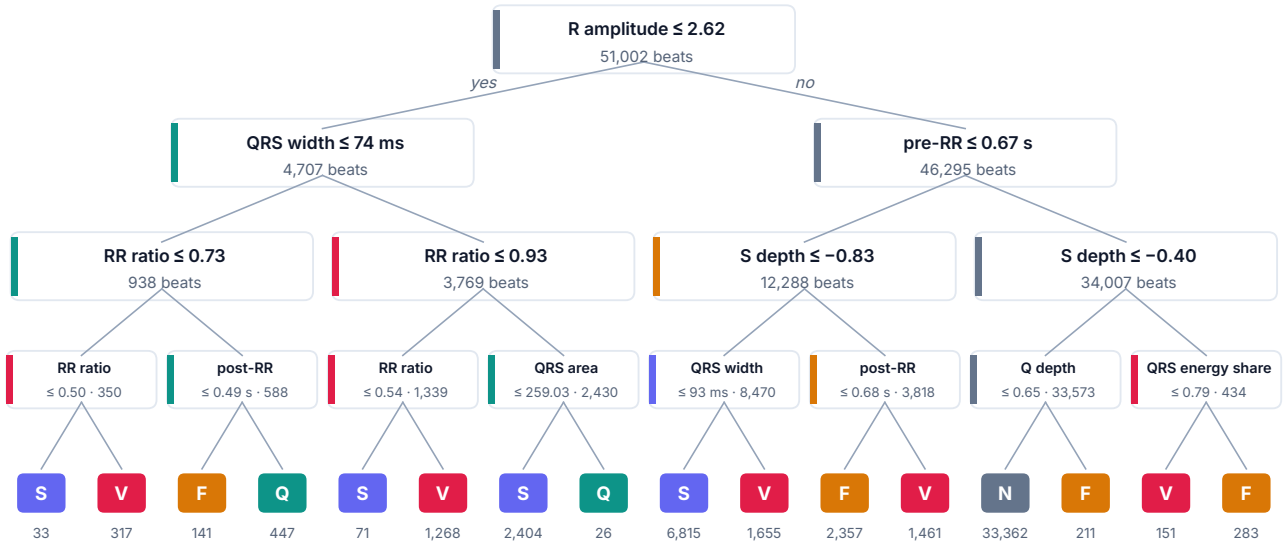


Figure 10. The learned tree (tree.json). Each box tests one feature; the left branch is taken when the condition holds. Colored bars and leaves show the class the tree would predict (with balanced class weights); numbers are training beats reaching the node. Amplitudes are in units of the normalized beat.

Features: QRS width (duration of the QRS complex); R amplitude (height of the R peak (normalized beat)); Q depth (deepest point just before R); S depth (deepest point just after R); QRS area (area under the QRS complex); T amplitude (height of the T wave); QRS energy share (share of the beat's energy inside the QRS); pre-RR (interval to the previous beat); post-RR (interval to the next beat); local RR (mean of the last up to 10 RR intervals); RR ratio (pre-RR / local RR (prematurity)).

5 · EVALUATION

Results on unseen patients (DS2)

All numbers in this chapter come from `metrics.json` (generated 2026-11-03): both models evaluated on the 22 test records of DS2 — 49,692 beats from patients the models never saw. The CNN result comes from an ensemble of 5 models trained with different random seeds; their class probabilities are averaged. Evaluated one by one, the 5 models reach a macro-F1 of 0.56 ± 0.05 (N, S, V, F).

Sensitivity (Se) = TP / (TP + FN): how many of the true beats of a class were found.

Positive predictive value (PPV, +P) = TP / (TP + FP): how many of the model's alarms for a class were right. **FPR** = FP / (FP + TN): false alarms among all other beats.

F1: harmonic mean of Se and PPV, high only if both are high. **Macro-F1** averages F1 over classes, so rare classes count as much as N.

Per-class results on DS2

		Sensitivity		PPV		F1	
Class	Test beats	CNN	Tree	CNN	Tree	CNN	Tree
N	44,240	**93.7%**	60.0%	**98.0%**	95.0%	**0.96**	0.74
S	1,837	**66.1%**	7.1%	**43.0%**	1.9%	**0.52**	0.03
V	3,220	**97.2%**	82.4%	**81.3%**	54.6%	**0.89**	0.66
F	388	0.8%	**1.8%**	**0.4%**	0.1%	**0.01**	0.00
Q	7	0.0%	**42.9%**	0.0%	**0.2%**	0.00	**0.00**
Macro-F1 (N, S, V, F)						**0.59**	**0.36**
Macro-F1 (all five)						0.47	0.29
Accuracy (supplementary)						92.1%	59.0%

The better of the two models is set in bold. Accuracy is listed only for completeness: with 89.0% normal beats it mostly measures the majority class.

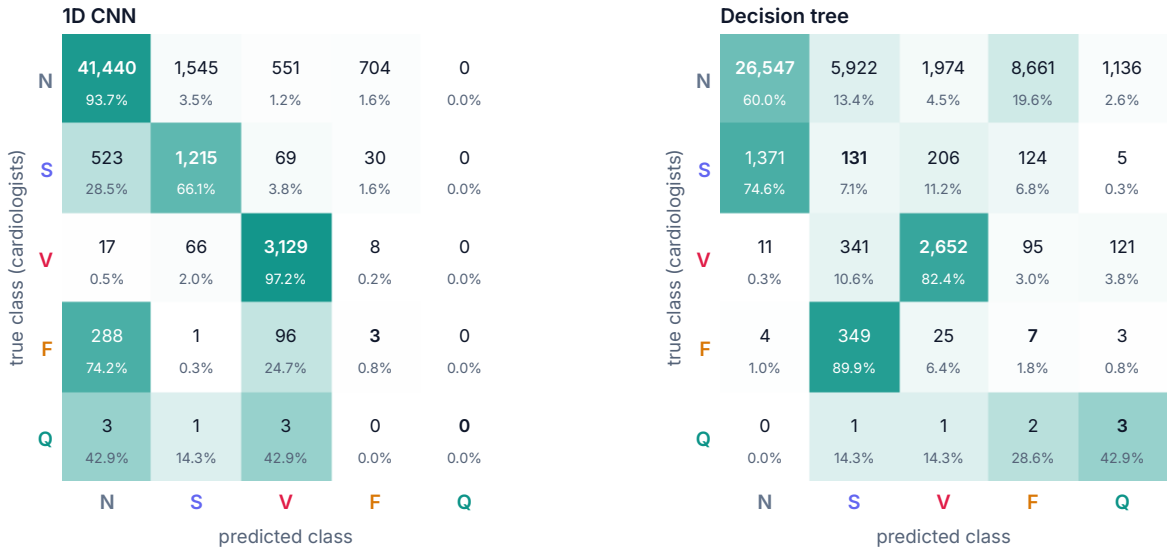


Figure 11. Confusion matrices on DS2: rows are the cardiologists' classes, columns the predictions; each row is colored by the share of its beats (percentages below the counts).

Reading the CNN's matrix

The three largest confusions are:

- 1,545 true N beats predicted as S (3.5% of all N beats)
- 704 true N beats predicted as F (1.6% of all N beats)
- 551 true N beats predicted as V (1.2% of all N beats)

The decision tree stays clearly below the CNN (macro-F1 0.36 vs. 0.59 on N, S, V, F): a handful of threshold rules on hand-made features is easy to read, but cannot capture the subtle shape differences the network learns.

Where the errors come from

Missed S beats are mostly called N (523 of 622 misses); false S alarms come mostly from true N beats (1,545 of 1,613). Missed V beats are mostly called S (66 of 91 misses); false V alarms come mostly from true N beats (551 of 719). Missed F beats are mostly called N (288 of 385 misses); false F alarms come mostly from true N beats (704 of 742).

Comparison with the literature baseline

De Chazal, O'Dwyer & Reilly (2004) introduced the DS1/DS2 protocol and reported, for their selected configuration on DS2, the values below (from the paper's abstract). It is the natural reference because it uses the same split and the same reference beat positions. A difference to keep in mind: their linear-discriminant classifier used hand-crafted morphology and interval features from *both* ECG leads, whereas this project uses one lead.

Class	Model	Se	+P	FPR
S	1D CNN (this project)	66.1%	43.0%	3.4%
S	Decision tree (this project)	7.1%	1.9%	13.8%
S	de Chazal et al. 2004	75.9%	38.5%	4.7%
V	1D CNN (this project)	97.2%	83.4%	1.3%
V	Decision tree (this project)	82.4%	54.9%	4.7%
V	de Chazal et al. 2004	77.7%	81.9%	1.2%

Values for this project are gross statistics (all test beats pooled), as in the paper; VEB: true F beats classified as V are not counted as V false positives.

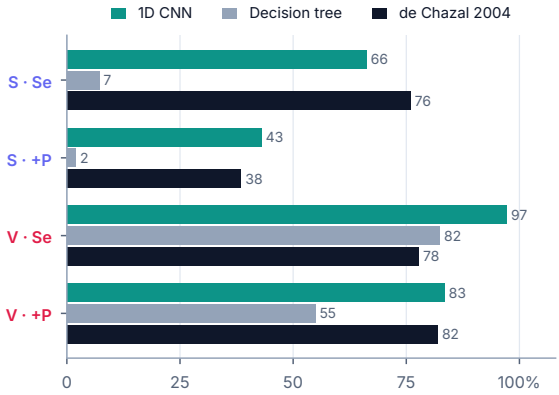


Figure 12. Sensitivity (Se) and positive predictivity (+P) for S and V beats, in percent.

Model selection on DS1 only (5-fold cross-validation)

Every design decision — training schedule, class weights, augmentation, the second lead, the seed ensemble and the class offsets — was made with record-grouped 5-fold cross-validation inside DS1 (3 seeds), never on DS2; experiments and decisions are logged in `docs/EXPERIMENTS.md` (plan steps A1–A8). In the “nested” row the offsets are fitted on the other folds only — an honest estimate of what they add. The table shows the selected configuration; on DS2 the final model reaches a macro-F1 of 0.592 (N, S, V, F).

Cross-validation on DS1	Macro-F1 (N, S, V, F)	S Se	S PPV	V Se	V PPV
Single models (mean ± std over 3 seeds)	0.481 ± 0.024	39.8 ± 13.4%	21.5 ± 3.8%	86.0 ± 4.2%	61.8 ± 11.1%
Seed ensemble	0.483	31.8%	22.0%	86.7%	61.7%
Ensemble + class offsets (nested)	0.461	14.8%	7.2%	73.4%	89.1%

End-to-end with Pan-Tompkins beats

The DS2 results above use the cardiologists' beat positions. A fully automatic system has to find the beats itself, so DS2 was also evaluated on the beats found by the Pan-Tompkins detector, with the same models. On the 22 DS2 records the detector finds 49,411 of 49,712 annotated beats (sensitivity 99.39%) with 106 false detections (PPV 99.79%; tolerance ±150 ms). Each detected beat takes the class of the nearest annotated beat within 150 ms; detections without one are left out of the scoring, and beats the detector misses cannot be classified at all — so read the table together with the detector's sensitivity.

	CNN · reference beats		CNN · detected beats	
Class	Se	PPV	Se	PPV
N	93.7%	98.0%	93.4%	97.8%
S	66.1%	43.0%	64.8%	41.3%
V	97.2%	81.3%	95.5%	80.5%
F	0.8%	0.4%	1.0%	0.5%
Q	0.0%	0.0%	0.0%	0.0%
Macro-F1 (N, S, V, F)		**0.59**		**0.59**
Macro-F1 (all five)		0.47		0.47

49,391 detected beats scored vs. 49,692 reference beats. Decision tree: macro-F1 (N, S, V, F) 0.36 on reference beats, 0.36 on detected beats.

External test: MIT-BIH Supraventricular Arrhythmia Database (no retraining)

How well does the model transfer to recordings it was never developed on? The final ensemble of 5 CNNs was applied unchanged — no retraining, no re-tuning of thresholds or offsets — to 78 records of the MIT-BIH Supraventricular Arrhythmia Database. These were sampled at 128 Hz and are resampled to 360 Hz before the same pipeline runs; the database does not say which lead its channel ECG1 is, so it may differ from MLII. Patients, recording devices and annotators differ from MIT-BIH, so this is a robustness check rather than a like-for-like comparison with the DS2 numbers.

		1D CNN on the external database		
Class	Beats	Se	PPV	F1
N	162,276	95.4%	97.3%	0.96
S	12,196	46.0%	60.5%	0.52
V	9,941	90.3%	67.0%	0.77
F	23	0.0%	0.0%	0.00
Q	79	0.0%	0.0%	0.00
Macro-F1 (N, S, V, F)				**0.56**
Macro-F1 (all five)				0.45

6 · LIMITATIONS

Limitations and honest methodology

A result is only as useful as its honest limits. These are the ones that matter most here.

The test set was looked at more than once

DS2 should ideally be evaluated exactly once. In this project it was evaluated twice during development (run 1: validation records 114, 122, 201, 208, macro-F1 over five classes 0.447; run 2: validation records 114, 119, 122, 201, macro-F1 over five classes 0.463; see `docs/EXPERIMENTS.md`). Between the runs, the choice of validation records was changed: record 208 holds about 90% of all DS1 fusion beats and was moved from validation back into training. The reason was a property of DS1, but the change was made after a DS2 result was known, so a small optimistic bias cannot be ruled out. After that, DS2 was locked: every further decision (plan steps A1–A8 in `docs/EXPERIMENTS.md`) was made by record-grouped cross-validation on DS1 with a selection criterion fixed in advance. The results in this report are the third and last evaluation of DS2, run exactly once with the frozen configuration and reported as they came out.

F and Q are practically not learnable here

DS1 contains 414 fusion beats, about 90% of them from one patient (record 208), and only 8 Q beats; DS2 has 7 Q beats. The CNN finds 0.8% of the 388 F beats in DS2 and 0.0% of the Q beats; its F alarms are right in 0.4% of cases. With so few, one-sided examples, a model cannot learn a pattern that transfers to new patients — larger and more diverse datasets would be needed.

Supraventricular beats (S)

The CNN finds 66.1% of all S beats, but only 43.0% of its S alarms are correct. S beats often look like normal beats and are mainly recognized by their timing, which also varies naturally in some patients.

Reference beat positions

The main results use the cardiologists' beat positions, the usual protocol for comparability. End-to-end on Pan-Tompkins beats (chapter 5) the CNN reaches a macro-F1 (N, S, V, F) of 0.59 instead of 0.59; beats the detector misses are not scored there.

Small, old, single-source data

44 usable records from one hospital, recorded in the late 1970s with Holter devices, one lead used. The only check on other data is the external test on the MIT-BIH Supraventricular Arrhythmia Database (chapter 5): macro-F1 0.56 (N, S, V, F); modern wearables were not tested.

Run-to-run variation

The reported CNN averages 5 models trained with different seeds, which smooths out part of the run-to-run variation; the single models alone reach a macro-F1 of 0.56 ± 0.05 (N, S, V, F). Differences of a few points between variants should not be over-interpreted.

Not a medical device

This is a research and portfolio project. It has no clinical validation and no regulatory approval, and it must not be used to diagnose or treat anyone.

What was done to keep the evaluation honest

- Strict inter-patient split; no record in both DS1 and DS2 (the only shared person, 201/202, is disclosed).
- Per-class sensitivity, PPV and F1 as the main metrics; accuracy only as a side note.
- Every design decision on DS1 only, by a selection rule (+1 standard deviation in cross-validation, N sensitivity not lower) written down before the main experiments. Caveat: it was set after a first CV round, whose stricter rule the chosen features would narrowly have failed.
- All experiments logged in `docs/EXPERIMENTS.md`; the web interface never shows placeholder numbers.

7 · PROCESS

How it was built: orchestrated AI agents

The project was planned and implemented with orchestrated AI agents. The point was not to let a model “write some code”, but to run software development like a small team: clear ownership, binding contracts, a shared status file and independent review.

Human (Leon Rohrer) — sets the goal and the ground rules (inter-patient evaluation, per-class metrics, no invented numbers), approves plans and anything that touches the test set or goes public, reviews results and texts.

↓

Mother agent (orchestrator) — writes the master plan, cuts the work into 18 modules, reads `project_state.json` before every wave and starts only modules whose dependencies are done (at most 16 sub-agents at once).

↓

Wave 1 · 13 agents in parallel on disjoint files

Data pipeline · 4	ML model · 4	Web & UI · 5
loader, filters, R peaks, segmentation	1D CNN, decision tree, training, evaluation	API, shell, diagrams, charts, content

↓

Wave 2 · integration — end-to-end pipeline CLI, then the web UI on real data.

↓

Wave 3 · independent review — three reviewer agents with different angles (ML correctness and leakage, tests, UI, honesty of claims); 17 findings, all reproduced and fixed by a fix agent.

↓

Wave 4 · full training DS1 → DS2 — blocked in the cloud sandbox (no access to PhysioNet), completed locally on Windows.

Status protocol
Agents never edit the status file directly. They report through `scripts/state.py`, which locks the file:
`start` · `done` (summary, decisions, files, tests) · `fail` (log excerpt → corrected restart) · `log` (deviation from a contract)

Interface contracts
`INTERFACES.md` fixes every function signature and data format up front, so modules can be built in parallel against each other.

Ownership
Each module owns its files exclusively; nobody edits another agent's files.

Tests
Every module ships with its own tests; the suite now has 222 tests. Agents report test commands and results with `done`.

Why this is more than a gimmick

Parallel agents only work if the problem is decomposed well. The contracts in `INTERFACES.md` made 13 agents productive at the same time without stepping on each other; the lock-safe status file made progress, decisions and failures traceable; and a separate review wave caught real bugs (for example a too-narrow QRS width measurement and uncapped class weights) before any result was published. The human role shifted from typing code to **specifying, deciding and verifying** — the skills that matter when automating real work.

Tech stack

Signal & ML	Python · NumPy, SciPy, wfdb, PyTorch, scikit-learn (python 3.13.5, numpy 2.5.4, scipy 1.18.1, torch 2.15.0, scikit-learn 1.9.1, wfdb 4.3.1)
Pipeline & API	command-line pipeline (`python -m ekg.pipeline`), FastAPI server, Makefile
Web interface	Vite, Tailwind CSS v4, Mermaid diagrams, hand-written SVG charts; German and English; hosted at leonrohrer.at
Quality	**222 automated tests** (counted with `pytest --collect-only` when this report was built); browser checks of the UI with Playwright; reproducible versions in `constraints.txt`
This report	generated by `docs/report/build_report.py` from the same JSON exports (matplotlib, HTML/CSS, Chromium)

EXPLORE IT YOURSELF

Every step, interactive

The web version shows the pipeline with live charts on real data, the rhythm player, both models and the agent status — in English and German.



leonrohrer.at/ekg-arrhythmia/

Live page	https://leonrohrer.at/ekg-arrhythmia/
Source code	github.com/LeonTheGoatRohrer/ekg-arrhythmia
Author	Leon Rohrer

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